GIZMO+ Source Verification Audit

Final pre-handoff verification • 12 September 2026

Purpose: verify that the GIZMO+ blueprint and traceability specification are grounded in the live Airtable schema, official product documentation, published packages, and authoritative music-rights sources. This document also identifies version-sensitive items and claims that remain design decisions rather than source facts.

1. Handoff Verdict

PASS WITH VERSION PINNING. The core architecture is supported by first-party sources. The handoff should instruct the implementation agent to pin compatible package versions, run package-level smoke tests, and treat the proposed GIZMO data model and UX conventions as architectural decisions derived from the verified legacy product—not as claims made by the external vendors.

2. Verified Source Matrix

Subject	Status	Primary source	Implication for GIZMO
Legacy GIZMO schema	LIVE / VERIFIED	Airtable GIZMO+ base appbeH0hdkwTdCfv6	8 tables and field/formula relationships re-read directly from Airtable.
Better Auth organizations	VERIFIED	better-auth.com/docs/plugins/organization	Organizations, members, invitations, active organization, roles and custom permissions are officially supported.
Better Auth + SQLite/Drizzle	VERIFIED	better-auth.com/docs/adapters/drizzle and /adapters/sqlite	Official Drizzle adapter supports provider='sqlite'; Better Auth CLI supports schema generation/migrations.
Turso database-per-tenant	VERIFIED	docs.turso.tech/api-reference/introduction	Platform API explicitly documents database-per-user / database-per-agent multi-tenancy and scoped tokens.
Turso Multi-DB Schemas	VERIFIED DEPRECATED	docs.turso.tech/features/multi-db-schemas	Do not use for a new implementation; the feature is deprecated for new users.
Turso engine choice	VERSION-SENSITIVE	docs.turso.tech/sdk/introduction	Current docs distinguish Turso Database from libSQL. New local/offline/sync work should evaluate @tursodatabase/database /Turso Sync; remote

			libSQL remains supported.
Mastra agent framework	VERIFIED	mastra.ai	Agents, typed tools, workflows, memory, observability/evals and durable workflow state are first-class framework capabilities.
Mastra + libSQL storage	VERIFIED	mastra.ai + @mastra/libsql package	LibSQLStore is officially supported for Mastra storage. Keep Mastra runtime state logically separate from GIZMO domain data.
Mastra + CopilotKit	VERIFIED	docs.copilotkit.ai/mastra and Mastra/CopilotKit official material	Current integration is AG-UI based and supports generative UI, shared state, human-in-the-loop and streaming.
AG-UI role	VERIFIED	docs.copilotkit.ai/ag-ui	AG-UI is the bidirectional agent↔️user protocol carrying state, tool calls, UI intents and interactions.
A2UI role	VERIFIED	docs.copilotkit.ai/mastra/generative-ui/a2ui	A2UI is a declarative generative-UI specification layered over the agent connection; it does not replace AG-UI.
CopilotKit Channels	VERIFIED	docs.copilotkit.ai/reference/channels	Managed Slack/Teams and direct Slack/Teams/Discord/Telegram/WhatsApp adapters are documented.
AIMock	VERIFIED	aimock.copilotkit.dev + @copilotkit/aimock	Mastra fixtures, deterministic LLM/tool tests, AGUIMock frontend testing and CI usage are explicitly documented.
ISRC semantics	VERIFIED	isrc.ifpi.org	ISRC identifies sound recordings/music videos—not musical compositions.
ISWC semantics	VERIFIED	iswc.org	ISWC identifies musical works; preserve separation from recording identifiers.
IPI semantics	VERIFIED	cisac.org + BMI creator FAQ	IPI identifies an interested party/natural or legal entity. BMI notes that multiple names/AKAs can have distinct IPI numbers.

Composition vs recording	VERIFIED	copyright.gov musician guidance and Circulars	The U.S. Copyright Office explicitly treats the musical work and sound recording as separate works.

3. Package / API Drift Register

Do not hard-code these versions into long-lived product documentation as timeless facts. They are a verified snapshot from the package registry on 12 September 2026 and should be rechecked when implementation starts.

Package	Verified snapshot / rule	Handoff instruction
better-auth	npm currently reports 1.7.x as latest in the version listing	Pin a tested version; generate the auth schema from the pinned CLI.
@mastra/core	npm snapshot: 1.66.0	Pin with the rest of the Mastra packages used by the project.
@mastra/libsql	npm snapshot: 1.22.x family	Verify compatibility with the pinned @mastra/core before install.
@ag-ui/mastra	npm snapshot: 1.1.2	Use current interrupt behavior and test it against the chosen CopilotKit runtime.
@copilotkit/runtime	npm snapshot: 1.71.0	Pin with frontend/runtime packages rather than using unbounded latest.
@copilotkit/channels	Official Channels docs show it must be installed as an exact tested pair with @copilotkit/runtime	Follow the pair published in current Channels docs at implementation time.
@copilotkit/aimock	Current package name is @copilotkit/aimock	Use deterministic fixtures in CI; retain real-model evals separately.

4. Corrections / Clarifications to Preserve

- Say “Turso / SQLite-compatible data layer” at the architectural level, then choose the concrete Turso engine/client after a startup spike. Current Turso docs distinguish the newer Turso Database engine from libSQL clients.
- Keep Better Auth control-plane data (users, sessions, organizations, memberships, invitations) separate from each workspace’s GIZMO domain database unless an implementation spike proves a different topology is preferable.
- AG-UI is the transport/interaction protocol. A2UI is an optional declarative UI mechanism. Controlled GIZMO components should remain the default for rights, destructive actions, and core business records.
- Mastra storage is agent/workflow/memory/observability infrastructure. It may use Turso/libSQL, but those tables are not the canonical music catalog.
- AIMock proves deterministic contracts and failure handling; it does not prove real-model quality. Maintain separate real-model eval datasets/scorers.
- ISRC belongs to the recording/master. ISWC belongs to the musical work/composition. IPI belongs to interested parties/names, not songs.

- The old Airtable PA/SR labels are legacy conveniences. In the modern model, store copyright registration/claim metadata in structures attached to the appropriate work type instead of encoding “PA” or “SR” as if each were a universal identifier type.

5. Still Not Source-Verified Because They Are Product Decisions

- Exact workspace database topology (one database per Better Auth organization is the current recommended GIZMO decision, not a requirement imposed by Better Auth or Turso).
- Exact ORM/query layer and migration runner.
- Object-storage provider for audio, artwork, agreements and generated PDFs.
- Email provider and tracking provider.
- The expanded pitch-status lifecycle beyond the four legacy Airtable statuses.
- Whether a pitch may contain one recording or multiple recordings at MVP.
- Whether offline/local-first Turso Sync is an MVP feature.
- Exact GIZMO custom roles and permission names beyond the Better Auth organization primitives.
- Final report/PDF rendering provider and template system.

6. Mandatory Pre-Code Verification Gates

Gate A — Package compatibility spike: Create a clean branch, install pinned Better Auth, Mastra, AG-UI/Mastra, CopilotKit Runtime/React, Channels and AIMock packages, and run their minimal examples together. Commit the lockfile.

Gate B — Auth/tenant spike: Create Better Auth user + organization + membership, resolve activeOrganizationId server-side, provision/select a Turso workspace database, and prove cross-workspace access is denied at the application service boundary.

Gate C — Agent/UI vertical slice: Seed Midnight Drive, ask “Show me Midnight Drive,” execute a typed Mastra read tool, stream over AG-UI, and render a controlled GIZMO Song/Recording component from real structured data.

Gate D — Mutation/HITL vertical slice: Ask to change a writer split, surface approval, authorize via workspace role, run a transaction, refresh UI, and write an audit event.

Gate E — AIMock regression: Reproduce Gates C/D with deterministic AIMock + AGUIMock fixtures and add failure cases (tool error, unauthorized mutation, malformed model call, interrupted approval).

Gate F — Real-model eval: Run the same representative prompts against the chosen production model(s) using Mastra scorers/evals. Do not replace this gate with mocks.

7. Source-of-Truth Hierarchy for the Coding Agent

1. Live GIZMO+ Airtable base — authoritative for legacy product behavior and legacy field/formula intent.

2. GIZMO Product & Technical Blueprint — authoritative for the target product architecture unless superseded by the traceability spec.

3. GIZMO Technical Traceability & Verification Specification — authoritative for requirement IDs, tests, verification state and implementation gates.

4. This Source Verification Audit — authoritative for external-source status, version-sensitive cautions and pre-code verification requirements.

5. Official vendor documentation and package registries — authoritative for current external API/package behavior; when they conflict with an older handoff example, current official docs win.

8. Final Handoff Instruction

The implementation agent must not silently reinterpret a requirement when an external package has changed. It must (1) identify the trace ID, (2) compare the pinned package/API to the official current documentation, (3) record any discrepancy as an architecture decision or blocked verification item, and (4) update tests before changing behavior.

Primary official sources used in this verification

- https://better-auth.com/docs/plugins/organization
- https://better-auth.com/docs/adapters/drizzle
- https://better-auth.com/docs/adapters/sqlite
- https://docs.turso.tech/api-reference/introduction
- https://docs.turso.tech/features/multi-db-schemas
- https://docs.turso.tech/sdk/introduction
- https://mastra.ai/
- https://docs.copilotkit.ai/mastra
- https://docs.copilotkit.ai/ag-ui
- https://docs.copilotkit.ai/mastra/generative-ui/a2ui
- https://docs.copilotkit.ai/reference/channels
- https://aimock.copilotkit.dev/
- https://aimock.copilotkit.dev/integrate-mastra/
- https://isrc.ifpi.org/
- https://www.iswc.org/
- https://www.cisac.org/
- https://www.bmi.com/creators/faqs
- https://www.copyright.gov/engage/musicians/
- https://www.copyright.gov/circs/